

Practical Laboratory 1

Two VM Network Commissioning and Connectivity Verification

WADF105 — Network Security Fundamentals

Full name	Mohammed Ibrahim
Student number	C11/26/FCDF/17175
Course	WADF105 — Network Security Fundamentals
Submission date	October 2026

Completion Questions

Question 29

What is the difference between the OPNsense WAN and LAN interfaces?

The WAN interface is the side of the firewall that connects to the internet. It gets its address automatically from VirtualBox NAT and is basically the firewall's door to the outside world. The LAN interface is the side that faces the internal network where the Ubuntu client sits. It has a fixed address of 10.10.10.1 and is what all internal devices use to get out to the internet. Traffic from inside goes to the LAN interface first, the firewall checks it, then sends it out through WAN.

Question 30

Why must both internal adapters use the same VirtualBox network name?

VirtualBox internal networks work like a virtual switch. If two adapters have the same name they end up on the same switch and can communicate with each other. If the names are different even by one character they are on completely separate switches and cannot communicate at all. That is why both the firewall LAN adapter and the Ubuntu adapter must both say exactly ICDFA-LAN with the same capitalisation, otherwise the two machines would be isolated from each other and none of the lab steps would work.

Question 31

What information does the default route provide to Ubuntu?

The default route tells Ubuntu where to send traffic when it does not have a specific path to the destination. In simple terms it is the address of the exit point for everything going outside the local network. Without it Ubuntu could only talk to devices on the same 10.10.10.0/24 network and nothing beyond that. With the default route pointing to 10.10.10.1, Ubuntu knows to send all outbound traffic to the firewall which then handles it from there.

Question 32

Which packet exchange allows Ubuntu to learn the firewall MAC address?

ARP, which stands for Address Resolution Protocol. Before Ubuntu can send any data to 10.10.10.1 it needs to know the physical MAC address of the firewall, not just its IP address. Ubuntu broadcasts an ARP request asking who has 10.10.10.1, the firewall responds with its MAC address, and Ubuntu stores that mapping so future communication can go directly to the right device. This exchange is visible in Wireshark when you apply the arp display filter and you can see the request and the reply clearly.

Question 33

Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

Pinging 1.1.1.1 uses a raw IP address so no name resolution is needed at all. DNS is a completely separate service that translates domain names like opnsense.org into IP addresses. If DNS was broken you could still reach 1.1.1.1 directly by its address but you would not be able to load any website by name. That is why the lab tests them separately, first ping an IP to confirm routing works, then resolve a domain name to confirm DNS works independently on top of that.

E7: Written Explanation

Why icdfa-nslab-client-v1 uses 10.10.10.1 as its default gateway

The Ubuntu client is on the 10.10.10.0/24 network and it only has one way out of that network which is through the firewall. The firewall's LAN interface sits at 10.10.10.1 and is the only router on this network. When Ubuntu needs to reach anything outside its own subnet, whether that is another network or the internet, it sends the traffic to 10.10.10.1 first. The firewall then checks the traffic and forwards it out through its WAN interface using NAT. Without 10.10.10.1 as the gateway Ubuntu would be completely stuck on the local network with no path to anything outside it.